INTRODUCTION

Ancient myths, religions, and philosophies all reflect an enduring fascination with how the world began and man's place in the story of life on Earth. In the West, Christianity held that all animals and plants were the result of a perfect creation. On the chain or ladder of being, no species could ever move from one position to another. Species were immutable, an idea called essentialism.

The 18th-century Age of Enlightenment began to challenge orthodox Christian beliefs. French zoologist Jean-Baptiste Lamarck rejected the prevailing Bible-based notion of Earth being only a few thousand years old. He argued that organisms must have changed from simple life forms to more complex ones over millions of years, and that the "transmutation" of species was the driving force behind this change. He speculated that characteristics acquired by animals during their lifetime were inherited by the next generation: giraffes, for example, became slightly longer-necked by stretching up to reach higher leaves, and passed this trait to their offspring; over many generations, giraffes grew longer and longer necks.

Fossil evidence of extinct life forms with features that resembled modern descendants, found by pioneering geologists such as Georges Cuvier, also suggested Earth had more ancient origins. Meanwhile James Hutton and Charles Lyell argued that geological features could be accounted for by the constant, ongoing processes of erosion, and deposition—a view called uniformitarianism. Because these processes take place slowly, Earth's history had to be much longer than was previously thought.